Agentic Swarm Discovery of Single-Site Water Molecule

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A compact water potential must reconcile liquid structure with pressure, density, response properties, and computational cost. We investigate an agent-assisted symbolic search in which each water molecule is represented by one neutral bead. A coordinator links function proposals and molecular-dynamics fitting, native force implementation, and independent validation through an auditable LLM-SR experience buffer. The selected Hamiltonian combines a refitted piecewise-cubic radial interaction, a tetrahedral angular invariant, a quartic coordination embedding, and a sparse low-coordination correction. Thirty-five distinct parameter sets were examined using 7.04 ns of training integration. After a disclosed training-only selection amendment and model freeze, 600 ps of fresh 512-bead simulations yielded density 0.99542 ± 0.00094 g cm^{-3} and EOS compressibility 4.435×10^{-5} bar^{-1}, compared with water references 0.99705 and 4.525×10^{-5}, respectively. The matched-temperature experimental radial-distribution RMS error is 0.12371, versus 0.13182 for native mW. All broad primary mean-value research gates pass, but pressure is sensitive to the EOS representation and thermal-response intervals include zero. Exact implementation optimizations preserve energies, forces, and the full global virial; measured runtime does not establish an advantage over native mW. The result is an inspectable single-site mechanical prototype and a reproducible discovery-and-diagnosis workflow, rather than a universally validated water model.

I. INTRODUCTION

Water is a demanding test of coarse graining because local structure and thermodynamic response must be described together. Removing explicit hydrogens eliminates orientational degrees of freedom and much of the molecular workload, but changes the observables the model can represent. A single-site model here means one neutral bead of mass 18.01528 u per water molecule. The bead is a coarse-grained molecular representation rather than a resolved molecular geometry.

Charge-free monatomic water is an established strategy. The mW potential encourages tetrahedral organization through short-range angular interactions [1]; subsequent machine-learning work has optimized coarse-grained water interactions [2]. We investigate an auditable agent-assisted search and qualification procedure, not the first water model without Coulomb interactions. The multi-property perspective of Izadi *et al.* [3] provides a methodological reference, while their explicit-site electrostatic representation remains distinct from the present model.

LLM-SR represents equation hypotheses as programs and separates structural proposals from numerical coefficient estimation [4]. We adapt that separation to expensive MD-in-the-loop fitting. Agents suggest functions and interpret failures; explicit Hamiltonians, measured trajectories, and recorded evaluation rules determine the evidence. The questions are whether independent radial, angular, and coordination terms improve multi-property balance, whether a training-selected model survives new simulations, and whether polynomial representations reduce runtime against an optimized short-range reference.

II. COLLABORATIVE SEARCH AND SELECTION

The recent campaign used a coordinator and three specialist contexts (Fig. 1). The fitting specialist proposed terms, ran temperature–density screens, and interpreted tradeoffs. The implementation specialist maintained analytic forces and virials, checked mechanics, and measured cost. The validation specialist prepared reserved seeds and evaluated the frozen candidate. These contexts shared structured messages and files; they were not independent human investigators or independently trained scientific models.

Actual upstream LLM-SR Sampler, Evaluator, ExperienceBuffer, and code-manipulation modules were used. A file-continuation adapter registered agent-authored hypotheses and hash-verified measured MD fitness. Agents inspected seven intermediate feedback snapshots before adaptive proposals. Numerical fitting and MD occurred outside LLM-SR. The final buffer records 40 evaluations, including repeated parameter sets at different sizes and budgets. This is a bounded agent-in-the-loop adaptation, not an unattended external LLM API search or proof of a global symbolic optimum.

Two seed families were used: a previous coordination-based prototype and a literature-informed polynomial approximation of mW. This round contains no new ab-initio energy or force training. The search varied radial shell corrections, positive angular-strength functions, stabilized coordination polynomials, coordination support, and undercoordination corrections. Training comprised 32 five-state 64-bead screens and eight six-state 512-bead refinements, representing 35 distinct parameter sets. Each small-cell state used 5 ps equilibration and 20 ps production; each refinement state used 10 ps and 30 ps, including a separate 295.15 K RDF state. The

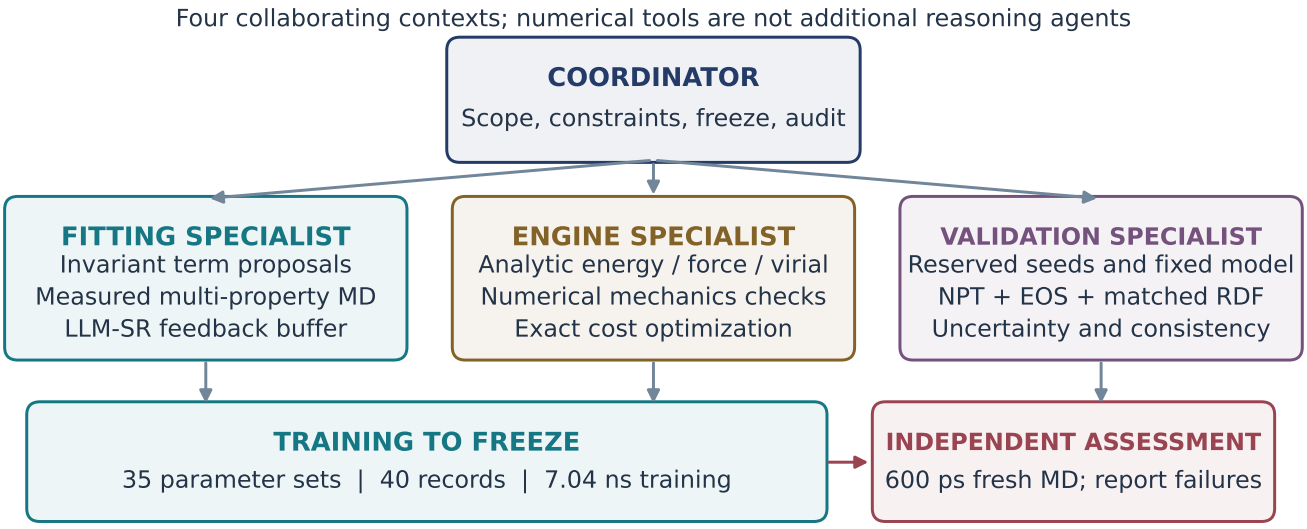


FIG. 1. Actual collaboration and evidence flow. The coordinator connects fitting, implementation, and validation specialist contexts. LLM-SR retains training fitness and feedback prompts. A model freeze separates training from fresh validation. Roles are not independent physical observables, and no controlled experiment establishes a causal swarm advantage.

retained training budget, including failed and excluded diagnostics, is 7.04 ns across 250 completed trajectories. Concurrent training loop time is not a controlled cost benchmark.

For property a, the discrepancy was $z_a = |y_a - y_a^{\mathrm{ref}}|/\tau_a$, with the largest discrepancy determining the primary score. Good agreement in one observable could not compensate for failure in another. Research tolerances were 0.5% density, 150 bar pressure intercept, 25% compressibility, 50% expansion and thermal-pressure coefficient, 2 kJ mol^{-1} cohesion-proxy error, 0.1 Å peak-position error, 30% peak-height error, and RDF RMS no larger than 0.15. These screening limits are much wider than experimental precision.

The original same-budget minimax winner was c33, score 0.93529. An explicit amendment after training, but before any fresh candidate trajectory, selected c34, score 0.97300: both its global-plane and central-anchored training density predictions lay within the existing density tolerance, whereas c33's did not. These were risk indicators, not NPT measurements, and short-run score differences did not establish statistical superiority. c33 is preserved as the original strict-score winner. The adaptive amendment is disclosed rather than described as an unchanged preregistered rule. c34 was frozen before testing; no held-out result replaced or retuned it.

III. INVARIANT SINGLE-SITE HAMILTONIAN

The selected potential is

$$U = \sum_{i<j} \phi(r_{ij}) + \sum_i \{F(d_i) + G(n_i) + \ell A_i\}, \quad (1)$$

$$d_i = n_i - n_0, \quad n_i = \sum_{j\neq i} \left[1 - r_{ij}^2/r_d^2\right]_+^3, \quad (2)$$

$$F(d) = K_1 d + K_2 d^2 + h_4 d^4, \quad (3)$$

$$G(n) = A_4 \left[1 - n/n_*\right]_+^4, \quad (4)$$

$$A_i = \sum_{j<k} w(r_{ij}) w(r_{ik}) (\hat{\mathbf{r}}_{ij} \cdot \hat{\mathbf{r}}_{ik} + 1/3)^2. \quad (5)$$

Here $[x]_+ = \max(0, x)$ and j, k denote neighbors of i. The frozen coefficients are

$$K_1 = -0.1667931559207833, \quad K_2 = -2.8,$$
$$h_4 = 0.2, \quad A_4 = 5.999564839230267,$$
$$\ell = 599.4640644, \quad n_0 = 2.536919437948505,$$
$$r_d = 5.5 \text{ Å}, \quad n_* = 0.5.$$

The five energy coefficients have units kJ mol^{-1}; coordination is dimensionless. The positive quartic stabilizes the negative quadratic embedding direction, without proving global thermodynamic stability.

The angular weight is $w(r) = t^3 Q(t)^2$, where $t = \left[1 - r^2/r_c^2\right]_+$, $Q(t) = \sum_{k=0}^{14} q_k t^k$, and $r_c = 4.3065$ Å. The exact q_k and pair coefficients are supplied in the Supplementary Information (SI) and machine-readable model. The pair term ϕ is piecewise cubic with physical support approximately 4.3065 Å and a quartic repulsive continuation below its first knot. Polynomial therefore means

piecewise radial polynomials and polynomials in invariant descriptors; distance and angular normalization still require square roots and division. This is not a single global Cartesian polynomial.

Positive functions $L(n) = \ell_0+\ell_1(n-n_{\mathrm{ref}})+\ell_2(n-n_{\mathrm{ref}})^2$ were also explored, with the $L'(n)A_i$ force included. The selected model has $\ell_1 = \ell_2 = 0$; the search does not establish that density-dependent angular strength is necessary. The sparse low-coordination correction provides a handle on the isolated-bead reference and sparse environments, whose interface consequences require separate tests.

The angular sum is evaluated exactly through neighbor moments. Let $s = \sum_j w_j$, $\mathbf{p} = \sum_j w_j\hat{\mathbf{r}}_j$, $\mathbf{M} = \sum_j w_j\hat{\mathbf{r}}_j\hat{\mathbf{r}}_j^{\mathrm{T}}$, and $s_2 = \sum_j w_j^2$. Then

$$A_i = \tfrac{1}{2}\{\mathbf{M}{:}\mathbf{M} + \tfrac{2}{3}|\mathbf{p}|^2 + \tfrac{1}{9}s^2 - \tfrac{16}{9}s_2\}. \tag{6}$$

This avoids explicit neighbor-triplet enumeration while retaining the angular interaction. Analytic forces and the global virial are implemented in a native LAMMPS plugin [5].

IV. INDEPENDENT MOLECULAR-DYNAMICS PROTOCOLS

Candidate-specific force differences, an isolated-bead energy audit, and 2 ps NVE at 1 fs preceded fresh validation. A fixed 10^{-6} relative-strain virial difference initially missed its numerical threshold. A step-convergence audit and two second-order Richardson estimates subsequently satisfied the unchanged absolute tolerance. The original failure and separate adjudication remain recorded; no Hamiltonian change was made. Full six-component virial checks were also performed for the engine and exact implementation aliases.

Fresh tests used 512 beads, a 1 fs timestep, reserved seeds, and the frozen model hash. NPT at 298.15 K and 1 bar used 30 ps equilibration plus 150 ps production. The EOS cross comprised densities 0.98, 0.99705, and 1.015 g cm^{-3} at 298.15 K, and temperatures 288.15 and 308.15 K at central density; each state used 10 ps equilibration plus 60 ps production. NVT Langevin sampling used damping 5 ps^{-1}. These static trajectories do not establish a physical diffusion clock.

The locked EOS estimator fitted actual mean temperatures:

$$P = P_0 + b(\rho - \rho_0) + c(T - T_0), \tag{7}$$

$$\kappa_T = (\rho_0 b)^{-1}, \qquad \alpha_P = c\kappa_T. \tag{8}$$

NPT volume/enthalpy fluctuations and canonical covariance responses were retained as independent consistency checks. A central-anchored plane and five-parameter quadratic cross diagnosed sensitivity without replacing the primary estimator. Five state means determine five quadratic coefficients exactly: there are zero residual degrees of freedom and no identifiable mixed density–temperature derivative.

The structural reference is the original numerical Skinner oxygen–oxygen curve [6] at 295.15 K, retrieved from a commit-pinned research mirror. A separate 10 ps equilibration plus 60 ps production run used that temperature and experimental density 0.997772886443 g cm^{-3}. Exact r^2-weighted shell averages were compared in 0.05 Å bins over 2.4–5.8 Å. Source-estimated errors were retained without assuming independent Gaussian bins. Thermodynamic references follow IAPWS [7]. Native stock Stillinger–Weber mW, rather than its polynomial approximation, was simulated with the same 600 ps static protocol. Old diffusion or phase results were not assigned to c34, and an earlier unmatched OPC3 RDF was not relabeled as experimental validation.

V. RESULTS AND UNRESOLVED RESPONSE

Density and compressibility central estimates are close to the water references (Table I). The candidate's matched RDF RMS is modestly smaller than native mW, including in the first shell. Its first peak is slightly too high and too far out, and intermediate-shell structure remains imperfect. It is not uniformly superior to mW: native mW's expansion central estimate is much closer to experiment.

All broad primary mean gates pass, with worst normalized discrepancy 0.93813. This does not establish full validation. The descriptive 95% paired-block-bootstrap compressibility interval is $[3.653, 5.500]\times10^{-5}$ bar^{-1}. Expansion is 1.366×10^{-4} K^{-1} with interval $[-3.333, 5.817]\times10^{-4}$ K^{-1}; the thermal-pressure derivative is 3.080 bar K^{-1} with interval $[-7.111, 12.950]$. Their signs and magnitudes remain unresolved. The maximum first/second-half pressure difference among the five states is 259.51 bar.

Pressure is a particular warning. The global-plane intercept is 66.45 bar, whereas the directly observed central state gives 182.65 ± 59.63 bar. Temperature correction gives 180.17 bar, and the saturated quadratic gives 181.19 bar. Global, anchored, and quadratic density roots are approximately 0.99416, 0.98908, and 0.99027 g cm^{-3}. This sensitivity prevents a qualified local-pressure claim. No favorable estimator is selected retrospectively to label pressure fixed.

NPT density is 0.995424 ± 0.000941 g cm^{-3}, with half means 0.994628 and 0.996221. Its compressibility agrees at present precision with the EOS result, while its expansion is also low and noisy. A density-only curvature correction can partly account for the pressure mismatch, but does not explain it completely. The exactly identified quadratic is a sensitivity calculation, not independent evidence of physical curvature.

The isolated-bead energy is $E_{\mathrm{iso}} = -3.3136661900$ kJ mol^{-1}, not zero. The proxy $E_{\mathrm{iso}} - \langle U\rangle/N + RT$ is 43.96791 ± 0.00559 kJ mol^{-1}. Its match does not es-

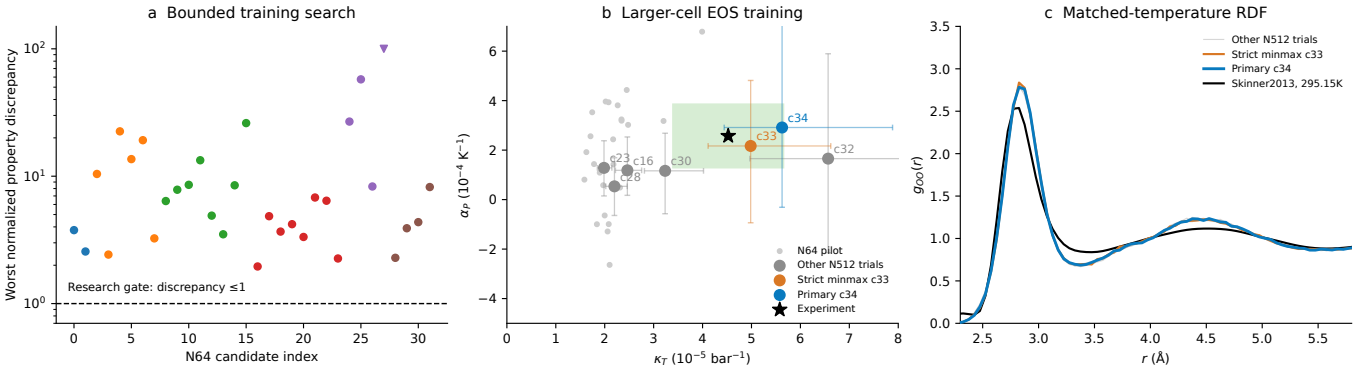


FIG. 2. Training search, larger-cell response tradeoffs, and matched-temperature RDFs. c33 and c34 are distinguished because the disclosed density-risk amendment selected c34 rather than the strict minimax winner. Response error bars are descriptive training intervals. The small-cell screen is not the final same-budget ranking.

TABLE I. Independent central estimates at ambient conditions, except matched 295.15 K RDFs. Density errors are descriptive block standard errors. Thermal EOS uncertainty is much broader than the displayed precision. Cohesion is an ideal-vapor proxy, not coexistence latent heat. Effective bead heat capacities are not complete molecular-water calorimetry. Full uncertainty and sensitivity results are in the SI.

Property	Previous candidate	Native mW	Frozen c34	Water reference
Density (g cm^{-3})	0.99990	0.99672	0.99542 ± 0.00094	0.99705
EOS κ_T (10^{-5} bar^{-1})	4.834	1.880	4.435	4.525
EOS α_P (10^{-4} K^{-1})	9.829	2.567	1.366	2.573
EOS $(\partial P/\partial T)_\rho$ (bar K^{-1})	20.334	13.654	3.080	5.686
Global-plane P_0 (bar)	−49.77	−16.58	66.45	1
Central-anchored P_0 (bar)	−57.37	−48.32	180.17	1
NPT mean pressure (bar)	−27.42	12.57	18.98 ± 7.62	1
NPT κ_T (10^{-5} bar^{-1})	4.772	1.841	4.111 ± 0.409	4.525
NPT α_P (10^{-4} K^{-1})	9.186	2.170	1.022 ± 0.730	2.573
Cohesion proxy (kJ mol^{-1})	43.954	44.647	43.968	43.987
Effective CG C_P (J mol^{-1} K^{-1})	28.79	31.26	36.02 ± 1.99	75.33
Effective CG C_V (J mol^{-1} K^{-1})	19.12	32.46	35.03 ± 1.29	74.54
Matched RDF RMS, 2.4–5.8 Å	—	0.13182	0.12371	0 by definition
Matched RDF RMS, 2.4–3.4 Å	—	0.23845	0.20938	0 by definition
Matched O–O peak position (Å)	—	2.825	2.875	2.800
Matched O–O peak height	—	2.081	2.767	2.570

tablish vapor pressure, boiling, or coexistence enthalpy because the sparse correction changes the gas reference. Effective CG C_P and C_V likewise omit molecular internal degrees of freedom.

VI. MEASURED COMPUTATIONAL COST

Eliminating electrostatics does not imply an advantage over an already short-range reference. We measured one-thread Apple M2 performance with 512 beads, common initial geometry, matched velocity seeds, 1 fs timesteps, 3 ps NVT preparation, and uninterrupted 10 ps NVE timing windows. Identical small native temperature/energy diagnostics were included; Python timestep callbacks and trajectory I/O were absent. Other campaign MD was stopped before timing.

A four-model, four-repeat counterbalanced series gave mean times 6.94 s for native mW, 4.99 s for polynomial mW, 11.02 s for primary c34, and 9.11 s for an exact shorter-cutoff alias. Removing identically zero radial tails reduced the neighbor cutoff from 6.0 to 5.5 Å without changing the physical interactions. Twenty-frame energies, forces, and all six global virial components were exactly unchanged, and short NVE checks passed. The polynomial control is faster in this workload; the conclusion does not transfer to the complete fitted Hamiltonian.

One bounded kernel optimization fused repeated coordination accumulation into a neighbor pass and skipped zero radial work. A separate three-model, three-repeat

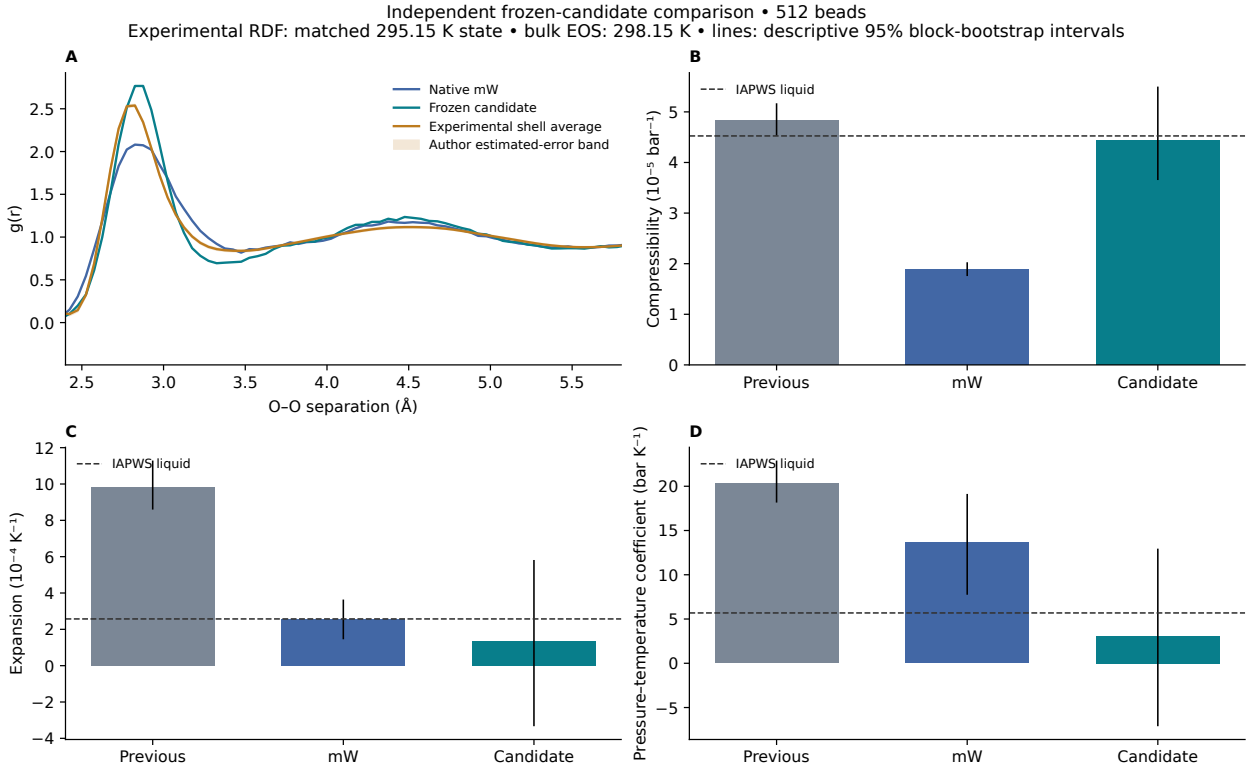


FIG. 3. Fresh structure and response comparison. The candidate improves compressibility relative to native mW and modestly reduces RDF RMS, while thermal-response intervals remain broad. Response intervals are descriptive block-bootstrap results, not coverage-certified errors. The RDF band is the source authors' estimated experimental error, not independent-bin likelihood weights.

series gave 7.586 s for native mW, 9.065 s for the existing alias, and 8.292 s for the fused kernel. Mean cost fell 8.53% relative to its matched alias control, but was 9.30% higher than native mW. Its 6.81–10.17 s range overlaps native mW's 7.41–7.79 s range, with substantial timing variability. No native-mW speed advantage is established. The two series are kept separate, and neither is an equal-accuracy, calibrated-clock, MPI/GPU, or production-timestep comparison.

VII. DISCUSSION AND NEXT INVESTIGATIONS

The failures are informative. Pair-well compensation for cohesion tended to stiffen the liquid, while large radial-coordinate warps caused pressure or structural failures. Broader coordination support provided a stronger compressibility handle, and the low-coordination correction partly separated gas-reference adjustment from bulk forces. These observations do not establish a unique or transferable functional form. A fast canonical/Born compressibility screen failed its EOS check and was excluded from selection. A manifest positional-lookup bug caused repeated-parent runs; these were preserved, excluded from intended-family ranking, and rerun with explicit identity checks.

The next priority is pressure consistency and thermal precision. Longer fixed-budget replicas of the central state and temperature pair, followed by independent intermediate-density states, should distinguish sampling from EOS curvature. A future training round could constrain direct state pressures and positive bulk modulus over a density interval while testing stabilized cubic embedding or independent angular-density response. A revised model would require another freeze and genuinely new validation.

Only after equilibrium evidence supports progression should a physical clock be calibrated jointly against diffusion and viscosity, with multiple sizes and uncertainty [8]. No c34 diffusion result is reported here. Density-versus-temperature/TMD, surface tension, vapor pressure, liquid–vapor and solid–liquid coexistence, gas clusters, and high-pressure stability remain separate studies. The neutral scalar bead has no molecular dipole observable or intrinsic dielectric relaxation under standard charge coupling; a field-consistent response variable would be an explicit extension.

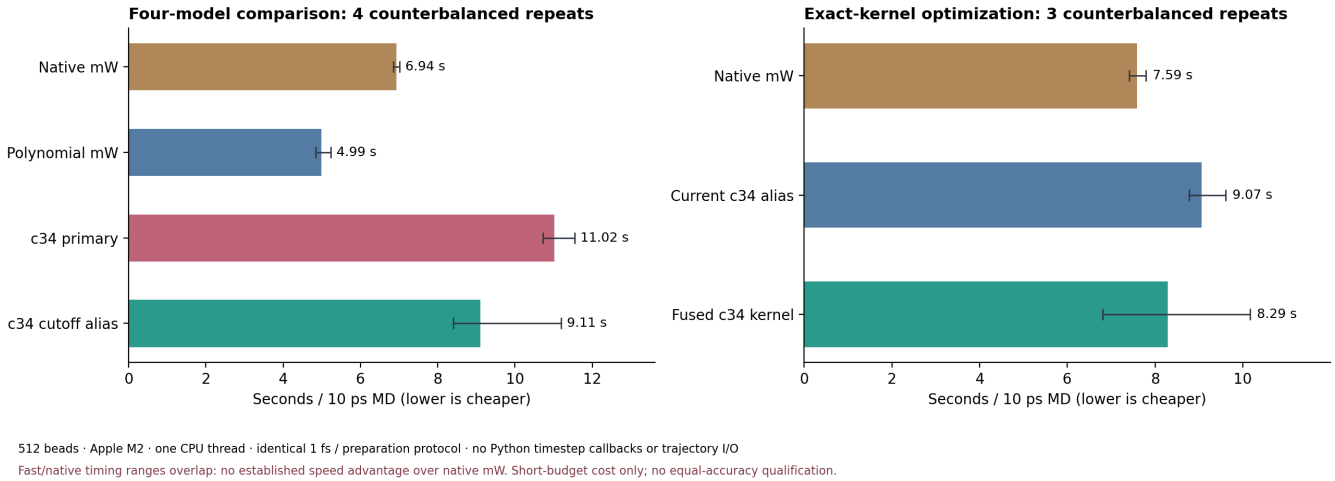


FIG. 4. Implementation-cost pilots. Whiskers show observed ranges, not confidence intervals. Exact optimizations reduce redundant work, but the complete candidate does not establish a native-mW speed advantage. The series have different repeat counts and are not pooled.

VIII. CONCLUSIONS

Agent-assisted symbolic search produced an invariant single-site prototype with encouraging density, compressibility, and RDF central estimates. Separating selection, mechanics, fresh validation, and cost measurement also exposed pressure sensitivity, broad thermal uncertainty, and the absence of an established native-mW speed advantage. The model remains unqualified for general-purpose water simulation. Its frozen expression, coefficients, negative trials, and exact implementation optimizations provide a concrete basis for further research.

DATA, CODE, AND AI-ASSISTANCE STATEMENT

The companion project includes frozen coefficients, native LAMMPS source, input snapshots, compact reports, experimental-reference provenance, and source hashes. Large raw trajectories remain in the campaign workspace and are not included in the compact Overleaf project. No public archival DOI is claimed. Language-model agents substantially assisted proposals, software, simulation orchestration, analysis, figures, and drafting. The author listing is provisional for review; no first-ever priority or causal swarm-performance benefit is established. Reproduction boundaries and the complete model are specified in the SI.

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